NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)

Official Study Material • Java Programming & OOP (JAVA101) • Unit 1: Java Fundamentals & OOP Concepts

Course Code:	JAVA101	**Regulation:**	R20 / R23 Autonomous
Subject:	Java Programming & OOP	**Unit:**	Unit 1 - Java Fundamentals & OOP Concepts

NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ACADEMIC STUDY MATERIAL: JAVA101 JAVA PROGRAMMING & OOP

UNIT 1: JAVA FUNDAMENTALS & OOP CONCEPTS

Regulation: R20 / R23 Autonomous Syllabus | Credits: 4

1. JAVA ARCHITECTURE & PLATFORM INDEPENDENCE:

Java is an object-oriented, class-based, platform-independent language developed by James Gosling at Sun Microsystems.

- Java Architecture:

* JDK (Java Development Kit): Full development environment containing compiler (javac), debugger, and JRE.

* JRE (Java Runtime Environment): Provides libraries, JVM, and core components to execute compiled Java programs.

* JVM (Java Virtual Machine): Abstract computing machine that executes compiled Java Bytecode (.class files).

* Bytecode & "Write Once, Run Anywhere" (WORA): javac compiles Java source code (.java) into platform-neutral Bytecode (.class). The host OS-specific JVM interprets and executes bytecode using JIT (Just-In-Time) compilation.

2. CORE OBJECT-ORIENTED PROGRAMMING (OOP) PRINCIPLES:

- Encapsulation: Binding data (fields) and methods into a single class unit and hiding internal state via private access modifiers.

- Abstraction: Displaying only essential features to users while hiding background implementation details.

- Inheritance: Mechanism where child class inherits attributes and methods of parent class for code reusability.

- Polymorphism: Ability of an entity to take multiple forms (Compile-time overloading and Runtime overriding).

3. BASIC JAVA SYNTAX & CONTROL STATEMENTS:

- Primitive Data Types: byte (1B), short (2B), int (4B), long (8B), float (4B), double (8B), char (2B, Unicode), boolean.

- Operators: Arithmetic (+, -, *, /, %), Relational (==, !=, <, >), Logical (&&, ||, !), Bitwise, Ternary (? :).

- Control Structures:

* Decision Making: if-else, nested-if, switch statement.

* Loops: for, while, do-while, enhanced for-each loop.

* Jump Statements: break, continue, return.

4. PRACTICAL CODE EXAMPLE:

```
// Java Program demonstrating basic OOP encapsulation, control flow, and arrays
public class StudentGradeDemo {
private String studentName;
private int[] marks;

public StudentGradeDemo(String name, int[] marks) {
this.studentName = name;
```

```
this.marks = marks;
}

public double calculateAverage() {
int sum = 0;
for (int mark : marks) {
sum += mark;
}
return (double) sum / marks.length;
}

public void displayReport() {
double avg = calculateAverage();
char grade = (avg >= 90) ? 'A' : (avg >= 75) ? 'B' : (avg >= 50) ? 'C' : 'F';
System.out.println("Student: " + studentName + " | Average: " + avg + " | Grade: " + grade);
}

public static void main(String[] args) {
int[] scores = {85, 92, 78, 88, 95};
StudentGradeDemo s = new StudentGradeDemo("Sita", scores);
s.displayReport();
}
}
```

Output:

Student: Sita | Average: 87.6 | Grade: B

5. IMPORTANT EXAM QUESTIONS & ANSWERS:

Q1: Explain JVM architecture and justify why Java is platform-independent.

Answer: JVM consists of ClassLoader, Bytecode Verifier, Memory Areas (Method Area, Heap, Java Stacks, PC Registers, Native Method Stacks), and Execution Engine (Interpreter + JIT Compiler). Java source code compiles into platform-independent Bytecode. Because different OS platforms run platform-specific JVM implementations that accept the identical bytecode, Java is platform-independent.

Q2: Differentiate between JDK, JRE, and JVM.

Answer: JVM is the runtime engine executing bytecode. JRE contains JVM plus standard class libraries. JDK is the software development kit containing JRE plus development tools (javac, javadoc, jar).